The Impact of Weather on Flight Delay

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Abstract

This is a sample article for illustrating how to write a reproducible research article using R and Quarto. preamble.tex is used to set up the LaTeX environment, such as packages and custom commands. title.tex (in the template folder) is used to customize the title page. american-journal-of-agricultural-economics.csl is the citation style for AJAE. It can be downloaded from https://github.com/citation-style-language/styles. The template folder contains these three files. bibliography.bib is the bibliography file

1 Introduction

Nothing is more frustrating than flight delays. The goal of this article is to understand what weather variables affect flights delays so that we can control weather when such time comes in the future[1].

2 Method

Regression analysis was used in this article. The econometric model writes as follows (Equation 1):

$$y_{i,t} = \beta_0 + \beta_1 temp_{i,t} + \beta_2 precip_{i,t} + \beta_3 wind_speed_{i,t} + v_{i,t} \tag{1}$$

2.1 Code availability

All computations (including creating tables and plots) were conducted using R software (R Core Team 2022), The Bergé (2018) package was used to run regressions.

[1]This is just for illustrating how to create a footnote

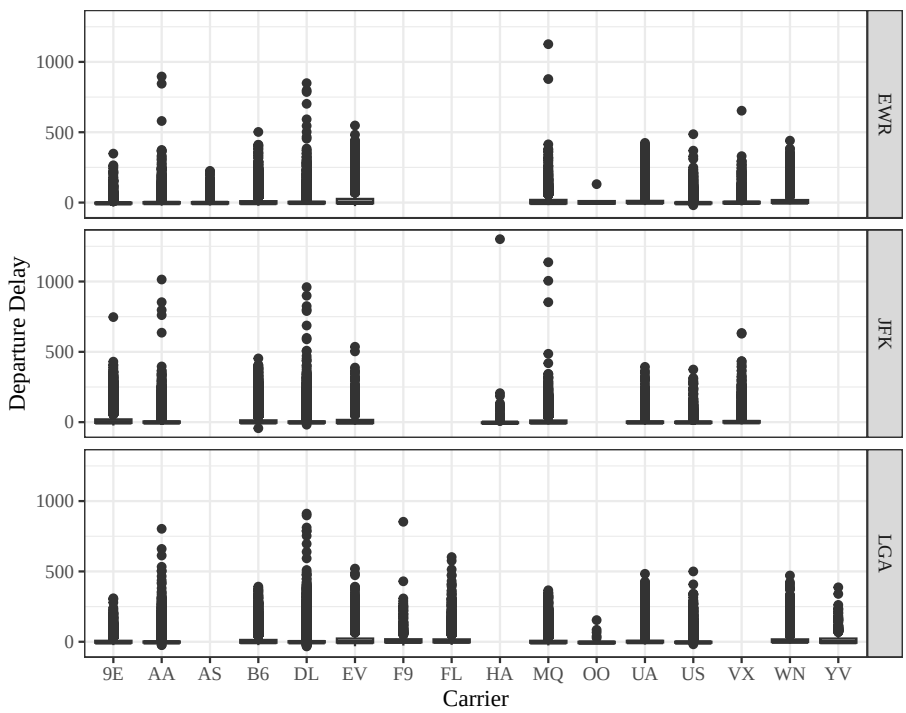
3 Data

We use publicly available datasets from the R nycflights13 package (Wickham 2021). Table 1 presents the summary statistics.

Table 1: Summary Statistics

	Mean	SD	Max
dep_delay	12.64	40.21	1301.00
temp	55.57	17.10	92.14
precip	0.00	0.01	0.15
wind_speed	10.45	4.19	56.39

Figure 1 shows the histogram of departure delay by carrier for each of the airports.



Note: The figure shows the distribution of departure delay by carrier for each of the airports.

Figure 1: Distribution of delay by carrier

4 Results and Discussions

Table 2 presents the regression results.

Table 2: Regression Results

	(1)	(2)	(3)
Temperature	0.145***	0.251***	0.251**
	(0.004)	(0.010)	(0.023)
Precipitation	379.641***	348.979***	348.979**
	(5.363)	(5.437)	(28.751)
Wind Speed	0.274***	0.326***	0.326*
	(0.018)	(0.018)	(0.059)
Num.Obs.	326 164	326 164	326 164
R2	0.020	0.034	0.034
Std.Errors	IID	IID	by: origin

+ p <0.1, * p <0.05, ** p <0.01, *** p <0.001

5 Conclusion

Weather matters.

References

Bergé, L. 2018. "Efficient estimation of maximum likelihood models with multiple fixed-effects: The R package FENmlm." CREA Discussion Papers (13).

R Core Team. 2022. R: A language and environment for statistical computing. Vienna, Austria: R Foundation for Statistical Computing. Available at: https://www.R-project.org/.

Wickham, H. 2021. nycflights13: Flights that departed NYC in 2013. Available at: https://CRAN.R-project.org/package=nycflights13.